Ameya S Moghe

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M.tech VLSI and Nanoelectronics
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Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
M.tech.	Indian Institute of Technology, Guwahati	8.67 (Current)	2025-Present
B.Tech.	Bhilai Institute of Technology Durg	8.96	2020-2024
Senior Secondary	CBSE Board	90.6%	2020
Secondary	CBSE Board	92.8%	2018

Experience

- **Indian Institute of Technology Guwahati** *Jul 2026 - Nov 2026*
  Teaching Assistant, Digital Circuits Lab *Jan 2026 - May 2026*
  Teaching Assistant, Control and Instrumentation Lab

Projects

- **Software Hardware Codesign For Image Transformer** *Jul 2026 - Present*
  Mtech Project under Prof. Dr. Sheikh Rafi Ahamed,Dept. of EEE IIT Guwahati
  – Dividing the Image Transformer architecture into software tasks and dedicated hardware blocks, and developing RTL code for the hardware blocks with data transmission tested via standard bus protocols.
  – Profiling the Image Transformer's attention and matrix-multiplication layers to identify the highest-impact stages for hardware offload.
- **AXI4-Lite Bidirectional Async-FIFO Bridge — RTL to GDS II** *Jul 2026 - Aug 2026*
  Project under Prof. Dr. Gaurav Trivedi,Dept. of EEE IIT Guwahati GitHub Repo
  – Designed a bidirectional AXI4-Lite asynchronous FIFO bridge with dual clock-domain crossing (CDC) using Gray-code pointers and 2-FF synchronizers in **Verilog HDL**.
  – Synthesized (**Cadence Genus**) and placed-and-routed (**Cadence Innovus**) on gpdk045 45nm CMOS across two asynchronous clock domains.
  – Achieved clean timing signoff — Setup WNS +2.633 ns, Hold WNS +0.049 ns, zero DRC/connectivity violations — independently cross-verified in **Cadence Tempus** (6 ps divergence, zero failing endpoints).
  – Characterized power in **Cadence Voltus** using real testbench switching activity — 1.402 mW total, dynamic-dominated (99.96%) with negligible static leakage (0.04%).
- **Design and Layout of 8bit Radix-4 Booth Multiplier** *Oct 2025 - Nov 2025*
  Course Project under Prof. Dr. Gaurav Trivedi,Dept. of EEE IIT Guwahati GitHub Repo
  – Designed and implemented 8-bit signed Radix-4 Booth multiplier in **Cadence Virtuoso** at the transistor level.
  – Validated output waveforms and timing behavior under different input combinations.
  – Developed full-custom layouts for all sub-modules using **Cadence Virtuoso**, ensuring DRC/LVS compliance via **Calibre** and verifying performance through post-layout **Spectre** simulations.
- **Design and Analysis of DA Based OBC 4 TAP Fir Filter** *Mar 2026 - Apr 2026*
  Course Project under Prof. Dr. Sheikh Rafi Ahamed,Dept. of EEE IIT Guwahati GitHub Repo
  – Designed a Distributed Arithmetic (DA) based FIR filter using Offset Binary Coding (OBC) to replace multipliers with a LUT-based shift-add architecture.
  – Implemented the design in **Verilog HDL** and simulated it in **AMD Vivado**, reducing hardware complexity and resource utilization compared to a direct-form multiplier-based implementation.

Technical Skills

- **Programming /HDL Language**: C/C++,Verilog
- **Scripting Language**: Bash*,Tcl* *(* elementary proficiency)*
- **Operating Systems**: Windows,Linux
- **Tools Used**: AMD Vivado,Cadence Genus,Innovus,Tempus,Voltus,Virtuoso(Schematic/Layout/ADE),Spectre Calibre(DRC/LVS),Synopsys VCS,Verdi
- **Language**: Hindi,English,Marathi

Key Courses Taken

- **Core**: Digital IC Design, Analog IC Design, System Hardware Design, Foundation of VLSI Architecture
- **Electives**: Advance Computer Architecture, Foundation of VLSI CAD

Achievements

- **State Level Basketball Player**
- **Visharad in Vocal Music** from Indrakala Sangeet Vishwa Vidyalaya Khairagarh (IKSVV)

Extracurricular and Leadership

- **Hobbies**: Basketball,Chess,Gardening,Cycling,Singing
- **Elected Department Placement Representative**: VLSI,EEE Department IIT Guwahati (May 2026 - May 2027)